JACKSONVILLE UNIVERSITY · HEALTH & EXERCISE SCIENCES

The Sport Science Decision Lab

A six-course pipeline for training Decision Intelligence in undergraduate sport science — built on a single architectural principle: students encounter the same data, the same reasoning framework, and the same professional role structure across every course. Only the stakes change.

The Lab exists to close the gap between analytical output and confident, defensible recommendations — what employers in sport science have named directly as the difference between a graduate who can run a test and one who can stand behind it. Six courses. One shared athlete database. One reasoning framework. One question, asked at every level: *can you explain your reasoning?*

Each course trains a distinct professional lens — data analyst, performance coach, performance analyst, athletic trainer, applied researcher, embedded practitioner — while the Decision Cycle and shared infrastructure thread the work together. AI is integrated from day one as a labor partner; what AI cannot replace, the student must own. The pipeline is organized into four tiers across six courses.

FOUNDATIONAL	FOUNDATIONAL	ACCELERATOR	CLINICAL	CAPSTONE	CAPSTONE
HLSC 205	**EXSC 325**	**EXSC 442**	**EXSC 405**	**HLSC 499**	**HLSC 490**
Data Analyst	*Performance Coach*	*Performance Analyst*	*Athletic Trainer*	*Applied Researcher*	*Practitioner*

THE DECISION CYCLE

One framework, every course

PHASE 01	PHASE 02	PHASE 03	PHASE 04	PHASE 05
Theorize	**Query**	**Interpret**	**Decide**	**Justify**
What is your model of how this system works?	What question are we asking, and of what data?	What does the output mean in context?	What is the recommendation, and what are the alternatives?	Why this call, defended under stakeholder pressure?

Theorize and Justify carry the most weight. AI now absorbs much of the middle three phases; what AI cannot inherit is the model the student commits to before the evidence, and what AI cannot compress is reasoning under live pressure from a stakeholder pushing back.

Dr. Chris Robertson · Associate Professor, Health & Exercise Sciences · Jacksonville University

SPORT SCIENCE DATA ANALYTICS

HLSC 205 Statistics for Health & Exercise Science

FOUNDATIONAL · 01

PROFESSIONAL LENS

Sport Scientist / Data Analyst

CORE QUESTION

"What does the data say?"

The entry point. Students encounter the shared sport science database for the first time — learning to query athlete data, select and execute statistical tests, and visualize results for stakeholders. Every analytical task is framed through the Decision Cycle: students do not simply run tests, they articulate why they chose a particular analysis, what it means in context, and what decision it supports. AI tools enter on day one as labor partners. Students own the reasoning.

ARCHITECTURE · FOUR UNITS + CAPSTONE

STATS 01 · Wk 1–4

Data Literacy & Description

Meeting the database. Variables, distributions, and learning to see before learning to test.

STATS 02 · Wk 5–7

Probability & Distributions

The mathematical scaffolding behind every inferential test that follows.

STATS 03 · Wk 8–10

Inference Foundations

Sampling distributions, CI, hypothesis testing, effect sizes — statistical vs. practical significance.

STATS 04 · Wk 10–12

Statistical Tests

One test per session: t-tests, ANOVA, correlation, regression, chi-square — and selection logic.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	Before touching the data, commit to a model: what do you expect athletes in this program to look like, and what would surprise you?
QUERY	Identify the decision and frame the stakeholder question. Choose the test that fits.
INTERPRET	Run the test; read output in context against your prior, not in isolation.
DECIDE	Translate the statistical output into a recommendation the Athletics Director can act on.
JUSTIFY	Defend test selection, assumptions, and the statistical-vs-practical significance call under questioning.

CAPSTONE

The JU Athletics Scouting Report

Commissioned by the Athletics Director. Pick a varsity program, query the database, select and defend the right test, and deliver a 1–2 page scouting report with presentation-quality visualizations. Deliverable: Scouting Report + Live Defense.

PERFORMANCE SCIENCE & TRAINING DESIGN

EXSC 325 Motor Control

FOUNDATIONAL · 02

PROFESSIONAL LENS

Performance Coach / Skill Acquisition Specialist

CORE QUESTION

"What should we do about it?"

The reasoning shifts from *description* to *prescription*. Same database, same athletes — but now the decision is a coaching prescription grounded in motor learning theory. Five notebooks build toward a Coaching Consultation: a motor control prescription for a specific athlete, defended live to the coach who has to implement it. A wrong answer here is not an inaccurate report; it is a practice plan that wastes a season or breaks an athlete's body.

ARCHITECTURE · FIVE NOTEBOOKS + CAPSTONE

MC 01 · Wk 1–2

Athlete Profiling

FMS, Y-Balance, baseline screens. Building the profile before designing anything.

MC 02 · Wk 3–5

Skill Acquisition

Fitts & Posner, learning curves, plateaus, variability — noise vs. exploration.

MC 03 · Wk 5–7

Practice & Load

Blocked vs. random, Challenge Point, ACWR — coaching prescription meets safety constraint.

MC 04 · Wk 7–9

Feedback Design

External focus, faded schedules, guidance hypothesis — restructuring how the athlete is coached.

MC 05 · Wk 10–12

Integrated Monitoring

Multi-table synthesis. The whole athlete, querying across all four prior notebooks.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	Hold a model of the athlete's motor system before reading the data. What stage of learning, what control mode, what would surprise you?
QUERY	The head coach brings a struggling athlete. What is the motor-control question buried inside the complaint?
INTERPRET	Read the athlete against theory — stage of learning, attentional focus, load, variability — before deciding the story.
DECIDE	Commit to a prescription: practice structure change, feedback schedule shift, or load intervention.
JUSTIFY	Defend the motor-learning rationale to a coach whose instincts may disagree.

CAPSTONE

The Coaching Consultation

A JU head coach commissions a motor control consultation for a struggling athlete. Build the analysis with Decision Cycle headers, produce a 1–2 page consultation the coach can read directly, and defend the prescription live to the coach who implements it. Deliverable: Coaching Consultation + Live Defense.

SPORT SCIENCE PREDICTIVE ANALYTICS

EXSC 442 Advanced Statistics — Predictive Analytics

ACCELERATOR · 03

PROFESSIONAL LENS

Sport Performance Analyst / Analytics Director track

CORE QUESTION

"What is likely to happen, and what should we build to anticipate it?"

The analytical accelerator. HLSC 205 teaches students to describe and test; EXSC 442 teaches them to predict, classify, and build the analytical systems sport science operations actually run on — injury prediction, performance forecasting, ensemble-based recommendations. The reasoning shifts from inference to prediction: the question is no longer "did this change help?" but "given what we know, what will happen next?" The output is a model. The competency is knowing which model to trust, and when.

ARCHITECTURE · SIX METHOD BLOCKS + CAPSTONE

MB 01

Logistic Regression

Probabilistic classification. Reading coefficients, calibrating thresholds.

MB 02

Mixed-Effects Models

Nested athlete data — within- vs. between-athlete signal extraction.

MB 03

KNN & Naive Bayes

Transparent classifiers exposing the bias-variance tradeoff.

MB 04

Trees → Forests → XGBoost

The ensemble ladder. Feature importance, SHAP, partial dependence.

MB 05

Evaluation & Validation

CV, ROC/PR, calibration — the deployment-readiness question.

MB 06

Integration

Multi-method convergent recommendation. Owning the disagreement.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	Before fitting anything, articulate the data-generating process you believe is at work. What features should matter, and why?
QUERY	Define the prediction or classification task precisely — what is being predicted, for whom, and with what operational tolerance?
INTERPRET	Fit, diagnose, and compare models. Read beyond accuracy: calibration, fairness across positions, failure modes.
DECIDE	Recommend a single model — or a convergent call across methods — and an operational threshold.
JUSTIFY	Defend the model choice, evaluation protocol, and deployment readiness against a skeptical performance analyst.

CAPSTONE

The Sport Science Intelligence Brief

A multi-method convergent recommendation. Students fit, evaluate, and compare models across methods, integrate the findings into a single brief, and defend the recommendation — and the model-selection call — under questioning. Deliverable: Intelligence Brief + Live Defense.

SPORTS MEDICINE & CLINICAL REASONING

EXSC 405 Advanced Anatomy — Clinical Neuroanatomy

CLINICAL · 04

PROFESSIONAL LENS

Athletic Trainer / Clinical Reasoner

CORE QUESTION

"What's injured, why does it matter, and when can this athlete return?"

The highest-stakes reasoning in the foundational tier. Every case begins with an athlete and an injury mechanism. Students use JU's Anatomage virtual dissection table to trace pathways, localize lesions, and predict deficits from the anatomy — structure by structure. In 205, a wrong answer is an inaccurate report. In 325, a flawed practice prescription. In 405, a wrong answer means a missed injury mechanism, an athlete sent back to play who shouldn't be — a sideline call that costs someone their season or their health.

ARCHITECTURE · FOUR UNITS + CAPSTONE

NA 01 · Wk 1–4

Spinal Cord & Pathways

The Anatomage. Thinking in pathways, not parts. Sports injury mechanisms drive every case.

NA 02 · Wk 5–7

Brainstem, CN, Cerebellum

Where pathways converge. Reasoning through multiple tracts simultaneously.

NA 03 · Wk 8–11

Basal Ganglia & Thalamus

From pathway tracing to circuit reasoning. Disorders as imbalances, not severed tracts.

NA 04 · Wk 12

Cortex & Integration

The launch pad for an unstudied capstone case. Extension beyond the syllabus.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	From the mechanism alone — before any imaging or detailed exam — what anatomy would you bet is involved, and what would surprise you?
QUERY	The athlete arrives with symptoms and a mechanism. What is the clinical question the trainer actually has to answer?
INTERPRET	Localize the lesion. Trace every affected pathway on the Anatomage. Predict every deficit the anatomy would produce.
DECIDE	Commit to a localization and a return-to-play / referral call.
JUSTIFY	Defend the localization against the best alternative explanation, including the family-facing version.

CAPSTONE

The Clinical Case Brief

A referring physician requests a neuroanatomical consultation on a case the student did not study in class. Trace the anatomy on the Anatomage, produce annotated screenshots, write the 2-page brief, and defend the localization against alternatives live.

Deliverable: Clinical Case Brief + Live Defense.

INDEPENDENT RESEARCH & ATHLETICS PARTNERSHIP

HLSC 499 Sport Science Senior Thesis

CAPSTONE · 05

PROFESSIONAL LENS

Sport Scientist / Applied Researcher

CORE QUESTION

"What does rigorous investigation reveal about a question our stakeholders actually care about?"

The simulation ends. Every prior course used a simulated database; in HLSC 499, athletes, coaches, and stakes all become authentic at once. Students develop, refine, and defend an argumentative senior thesis — and through the JU Athletics research partnership, that thesis is grounded in real institutional data answering a question a real coaching or training staff member wants answered. The schema and Decision Cycle are identical to the foundational tier. Only the data is real. This transition is the structural payoff of the entire pipeline.

ARCHITECTURE · EIGHT MILESTONES + LIVE DEFENSE

MS 01
Thesis Question
Argumentative, not topical. Screen-recorded reasoning.

MS 02
Prospectus
Where weak projects are caught. Counter-position required.

MS 03
Annotated Lit Review
Sources as evidence, not summary. NotebookLM workspace.

MS 04
Thesis Outline
The argumentative skeleton. Counterargument placement.

MS 05
Introduction
Setting the stakes. Stakeholder framing.

MS 06
Arguments Draft
The substantive core. Highest-weighted milestone.

MS 07
Conclusion
Synthesis + executive summary seed for the staff.

MS 08
Final + Defense
Live defense. Executive summary delivered to staff.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	Before the literature review, before the analysis, articulate your model of the system the staff is asking about. What position are you taking?
QUERY	A real stakeholder question, owned independently. What does the Athletics staff actually need to know?
INTERPRET	Real institutional data. Apply the 205/442 toolkit with stakes attached — methodology choices now affect a real program.
DECIDE	Commit to an argumentative claim. Name the strongest counterargument and hold position against it.
JUSTIFY	Defend the claim live. Deliver the executive summary to the coach who asked the question.

CAPSTONE

Argumentative Senior Thesis + Live Defense

The thesis advances a clear, defensible claim supported by evidence and reasoning. Athletics-partnership students produce a dual deliverable: the academic thesis and a practitioner-facing executive summary delivered to the staff who posed the question. Deliverable: Thesis + Executive Summary + Live Defense.

PROFESSIONAL PLACEMENT & APPLIED PRACTICE

HLSC 490 Sport Science Practicum

CAPSTONE · 06

PROFESSIONAL LENS

Sport Science Practitioner

CORE QUESTION

"How do I function competently in a real professional environment?"

Every prior course was preparation. In HLSC 490 the preparation ends. Students are placed with JU Athletics or external sport science partners as Sport Science Interns, applying the full Decision Cycle to real athletes, real coaches, and real operational pressure. There is no simulation left. The Decision Cycle becomes internalized professional practice — the way the student approaches every problem in a sport-science setting. Four placement models accommodate different student strengths and site needs.

ARCHITECTURE · FOUR PLACEMENT MODELS

MODEL A · 1–3 cr

Single-Sport Embedded

One varsity sport, full season. Practice, competition, data collection, AT shadowing — depth in one program.

MODEL B · 3 cr

Three-Lens Rotation

Analytics, S&C, Athletic Training — each foundational lens in sequence. Capstone is a multi-lens case study.

MODEL C · 2–3 cr

Project-Based

A real Athletics question. Proposal, analysis, staff presentation. Deliverable is a decision tool, not a paper.

MODEL D · 2 cr

Athlete Case Management

Longitudinal 3-lens dossier on 3–5 athletes. Performance, motor control, injury risk — across the semester.

DECISION CYCLE · LOCAL EXPRESSION

THEORIZE	Walk into the placement with a model of how this team, this program, this athlete operates. Update it daily. Defend it when challenged.
QUERY	The question comes from a coach or trainer under operational pressure. Translate it into something the data can answer this week.
INTERPRET	Real performance records, real injury tracking — applied against the frameworks built in simulation.
DECIDE	Recommendations that affect what happens in practice tomorrow, not next semester.
JUSTIFY	Defend to the professional who will implement — or reject — the recommendation.

CAPSTONE

Competency Demonstration in a Placement

Students apply the full Decision Cycle to real athletes, coaches, and operational pressure. Assessment is competency-based across five domains, evaluated by the on-site mentor. Deliverable: Portfolio + Exit Presentation.

THE CONNECTIVE TISSUE

One pipeline, one framework, one question.

The same question runs through every course in the Decision Lab: *can you explain your reasoning?* The content changes. The professional lens changes. The stakes change. The question does not.

The Decision Cycle in Five Phases

The Lab teaches reasoning, not procedure. Every course in the pipeline operates on the same five-phase cycle: **Theorize → Query → Interpret → Decide → Justify** — and the phases are not equally weighted. AI now absorbs much of the analytical pipeline; the middle three phases (Query, Interpret, Decide) are increasingly AI-assisted, and the work happens well in partnership with the student. What AI cannot do is theorize for the student: committing to a mental model before the evidence is the act of taking a position. What AI also cannot compress is justifying a recommendation under live pressure from a stakeholder pushing back. *Reasoning under pressure is the competency that does not compress, and reasoning before evidence is the competency that AI cannot inherit.* Students are evaluated not on whether they produced correct output but on whether they can walk through the cycle under questioning — including back to the model they committed to before they ran the analysis.

Five Operating Principles

PRINCIPLE 01

AI can assist. Only you can think.

AI is a labor partner. Reasoning, at both ends of the cycle, remains the student's.

PRINCIPLE 02

The only variable is the stakes.

Database, framework, role structure, and reasoning demand are constants. What changes is how real the consequences become.

PRINCIPLE 03

Competency is demonstrated, not declared.

Every course ends in live demonstration under questioning. Written outputs support the defense. They do not replace it.

PRINCIPLE 04

Stakeholder communication is the primary output.

The job is not to produce analysis. It is to produce a decision recommendation a non-technical stakeholder can act on.

PRINCIPLE 05

Prepared, not planned.

Practitioners cannot script the situations they will face. The Decision Cycle is the operational expression of preparedness.

SHARED INFRASTRUCTURE

The database, the cycle, the question.

A student who built a dashboard in 205 builds one with real data in 490 without learning a new system. The infrastructure is familiar. Only the stakes change.

What the Pipeline Produces

The Lab is built to produce a **transdisciplinary practitioner** — a sport scientist who reads any sport-science problem across the disciplines rather than from inside one of them. The pipeline is multidisciplinary scaffolding; the Decision Cycle and the shared athlete database are the interdisciplinary infrastructure that connects them; the graduate is what emerges from inhabiting every seat at the sport science team table on shared reasoning. Many programs claim transdisciplinary integration. Few have a structural mechanism for producing it.